

Leveraging Human-AI Partnership Approach to Unravel Human Psychological Constructs

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Abstract

Human psychological constructs form the cornerstone of our understanding of cognition, emotion, and behavior, which are crucial to health informatics and HCI studies. However, current psychological construct analyses may lack theoretical grounding, cultural sensitivity, context, and socio-digital tailored interventions. My thesis explores how these human-centered computing methods, in particular human-AI partnerships, can unravel psychological constructs, using mental health stigma as a case study. Specifically, the study takes a multi-faceted approach: designing conversational agents for multilingual, cross-national data collection from over 1,000 participants; conducting human-AI collaborative qualitative analysis to interpret the embodied psychological constructs; modeling causal relationships from human-chatbot conversations to decompose psychological factors and their interplay; dissecting cross-sociocultural variation; and exploring digital techniques for restructuring psychological constructs. The overall goal is to enhance human-centered AI approaches for psychological-construct analysis, foster cultural inclusivity, inform the design of culturally appropriate healthcare technologies, and promote social justice by operationalizing psychological constructs.

CCS Concepts

• **Applied computing** → **Psychology**; • **Human-centered computing** → **Empirical studies in HCI**.

Keywords

Human-centered AI, Psychological construct, Mental health stigma, Culture

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1 Background and Motivation

Psychological constructs are theoretical concepts created to explain observable patterns in human behavior and thinking [19], such as personality traits, emotions, attitudes, and beliefs, which may be inferred from languages. Such constructs are fundamental to understanding user behavior and mental processes. Unraveling these psychological constructs is crucial as it provides a theoretical foundation for understanding human-technology interaction and informs the development of digital interventions and treatments for a wide range of psychological issues. For instance, psychological constructs like politeness and stereotypes formed the foundation of the computers-are-social-actors theory, which revealed that people unconsciously treat computers as social beings [44]. However, their complex nature poses challenges for researchers - conventional *quantitative* measurement instruments have limitations, such as social desirability bias [45], insufficient user engagement [36], and overlooking the complex and dynamic nature of human psychological constructs [23], while *qualitative* assessment also faces issues of limited reach, replication difficulties [2], and labor-intensive analysis with substantial datasets [63].

Recent advances in AI and computational technologies have opened up new possibilities for studying psychological constructs. On the one hand, chatbots [25] offer a promising approach to psychological-construct *collection*. They can provide a middle ground between questionnaires and interviews, offering a more engaging experience [30] than the former while being more scalable than the latter. In particular, AI-powered chatbots are flexible, fast, and adaptable tools that can gather rich qualitative insights into complex social and psychological phenomena by combining gregarious interactivity with wide reach [69]. On the other hand, AI and computational tools like LIWC demonstrate powerful capabilities in *dissecting* psychological constructs - they can, for instance, quantify abstract constructs such as emotions, dispositions, and biases [51, 59], uncover cultural concept interconnections [38], assess real-time societal and emotional impacts during social events [55], and detect specific psychological phenomena like social stigma [18, 43] through automatic text analysis. These AI-enabled analyses potentially make psychological insights more accessible and scalable, helping democratize psychological knowledge and support evidence-based practices [53]. Large language models (LLMs) have also shown promise in delivering digital interventions by rapidly generating messages that promote healthy habits [28, 42, 65], augmenting clinical decision-making with evidence-based suggestions [58], and increasing the accuracy of mental health screening [34].

Despite these technological advances, a potential limitation persists - most current approaches are primarily **descriptive** rather

than **explanatory**. Tools like LIWC reduce complex psychological phenomena to simple word frequencies or statistical patterns, for instance, counting anger-related words to measure anger levels [7, 12, 50]. While AI shows promise in examining language for psychological constructs, such methods may lack theoretical grounding and face challenges in incorporating how language is socially constructed through cultural contexts, social interactions, and shared meanings [22, 57]. In contrast, taking anger as an example, a theoretically grounded approach would examine how language reveals a person's cognitive appraisals (e.g., perceived unfairness), behavioral tendencies (e.g., argumentative responses), and contextual triggers when experiencing anger [37]. As McCrae & Costa [40] argued, such explanatory analysis is crucial for unraveling human psychological constructs - their antecedents, components, and consequences.

In addition, much of the current AI-driven analysis of psychological constructs relies heavily on social media data, which comes with inherent limitations. Psychological constructs are often culturally specific and require consideration of socio-cultural nuances to be fully understood [20]. Social media data is largely **decontextualized** [11], making it difficult to account for individual traits, cultural backgrounds, and situational factors that may influence the expression of psychological constructs. The meanings derived from such data could be limited or even misleading when taken out of context [21].

Accordingly, **cross-cultural analysis** is essential for unraveling psychological constructs because culture partly shapes how individuals interpret and interact with the world through shared attributes, meanings, values, and beliefs [10]. For instance, studies have shown that Chinese populations hold particularly stigmatized attitudes toward mental illness based on core Confucian values [27, 33]. Different cultural groups thus exhibit distinct patterns in technology adoption, interface preferences, and digital interaction styles. However, current AI-assisted analysis predominantly reflects Western-centric culture [1, 16], with a particular bias toward English-language data from social media. Yu et al. [66] found that nearly two-thirds of hateful communication datasets are in English, with a heavy concentration of contributions from U.S.-based researchers. This linguistic and socio-cultural bias may limit AI's ability to serve users from diverse backgrounds, particularly in regions like Asia, where multiple languages and cultural practices remain underrepresented in AI models [46].

Lastly, while AI has shown promise in dissecting psychological constructs, there is untapped potential to leverage these obtained insights for developing **tailored and culturally aware digital interventions**. Research has shown that individuals prefer and are more easily persuaded by messages that match aspects of their self, such as their values, social identity, or gender [52]. Personalized content can foster "bridging experiences" [67] that increase individuals' engagement with the message and convey a sense of "me-ness" [52]. This personalization has been linked to increased empathy and prosocial responses [5, 6, 49]. Despite these known benefits, many existing digital intervention approaches still adopt a relatively one-size-fits-all strategy [17, 31, 54], which may overlook individual differences and cultural nuances. In addition, current tailored methods remain relatively rigid, typically limited to selecting treatment modalities, prescribing specific modules or exercises,

adjusting treatment intensity, or providing individualized feedback [47].

2 Research Objectives and Approach

To address the existing gaps in AI-driven psychological-construct analysis, including the lack of theory-driven explanatory approaches [12], decontextualized data-collection methods [11], insufficient cultural awareness [66], and the absence of flexibly tailored interventions [36], I employ a multi-faceted approach that integrates various HCI techniques with AI technologies. My thesis aims to leverage the complementary strengths of human expertise and AI capabilities to deepen our understanding of psychological phenomena, uncover cross-cultural nuances, and develop effective digital interventions. This research is crucial as it explores the transformative potential of AI in social psychology while using innovative HCI approaches to potentially revolutionize how we unravel psychological constructs. The primary research questions guiding my study are:

- **RQ1:** *How can human-AI partnerships advance theory-grounded explanatory psychological-construct analysis?*
- **RQ2:** *To what extent can AI-enabled technologies reveal cross-cultural variations in human psychological constructs?*
- **RQ3:** *How can insights derived from human-AI collaborative analysis be translated into tailored, culturally sensitive digital interventions?*

Specifically, by focusing on *mental health stigma* as a case study, for RQ1, I first designed AI-powered chatbots to simulate semi-structured interviews and collected conversational data from 1,002 participants. Then, I proposed a human-AI collaborative theory-driven qualitative analysis pipeline where both human and AI coders perform deductive coding based on existing theoretical frameworks. Their coding disagreements are then analyzed through inductive coding to derive new insights. To further unpack the processes, triggers, and mechanisms of stigma formation, I construct causal knowledge graphs that map interrelationships between psychological factors, informing the development of conceptual models and network analysis. In addition, I plan to develop an interactive visual analytics system [64] that renders these causal knowledge graphs and provides a platform for theory-grounded causal analysis.

For RQ2, the methodology involves multi-national data collection, focusing on Japan, Singapore, China, and the U.S.A. [48, 56], to capture a spectrum of cultural orientations from collectivism to individualism. I plan to develop multilingual chatbots, followed by a comparative analysis of linguistic markers, topic modeling, and causal relationships [62] across these diverse cultural contexts. This human-AI approach aims to reveal how different cultural orientations manifest in human-chatbot conversation, helping identify cultural variations in psychological constructs.

To address RQ3, I will design chatbots for delivering personalized interventions aimed at promoting social justice, particularly in the context of mental health equity. This approach will be grounded in the actionable knowledge gathered from RQ1 and RQ2, integrating principles of social contact, education, and therapeutic techniques. The interventions will be designed to challenge social stigma, promote inclusivity, and advocate for equitable treatment of individuals

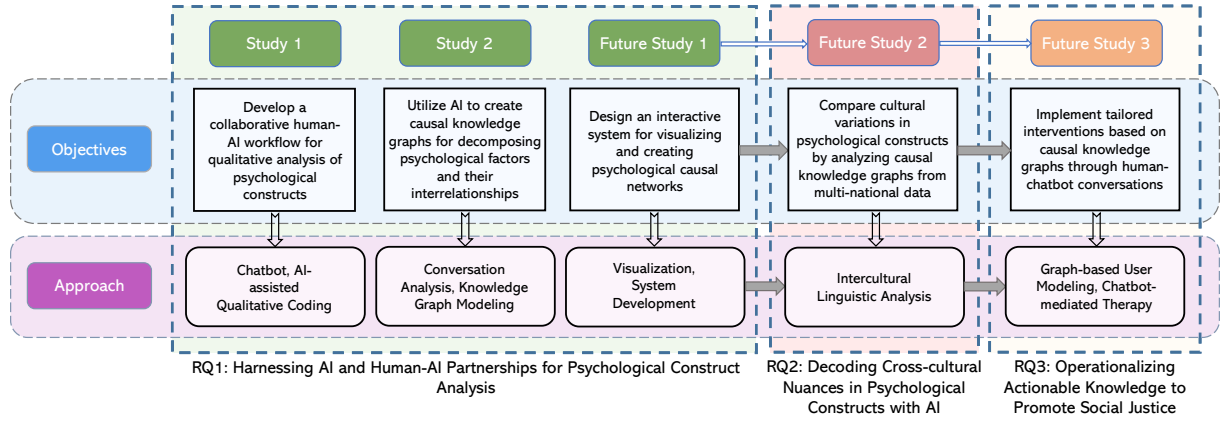


Figure 1: Overview of research to date and expected next steps to address research questions.

with mental health conditions. This involves developing knowledge retrieval and retrieval-augmented generation technologies to tailor content to individual users' cultural backgrounds and experiences [4, 47]. I will also explore strategies for technology-enhanced sustained engagement to facilitate long-term attitudinal and behavioral changes.

3 Results and Contributions to Date

3.1 Study 1: AI for Theory-grounded Psychological-construct Analysis

Previous research has explored LLMs' potential in theory-driven qualitative coding, (i.e., deductive coding) tasks, demonstrating their ability to replicate human coding with high consistency [26, 32, 63]. However, these studies primarily focused on LLMs as human coder *replacements*, overlooking their potential as collaborative partners in advancing theoretical understanding. Additionally, while some studies have used LLMs to clarify code definitions *within* existing theories [14, 68], their potential to extend *beyond* existing theories remains largely unexplored, particularly in the context of socio-psychological phenomena like mental health stigma. We developed CHALET (Collaborative Human-AI AnaLysis for Empowering Theory-driven qualitative coding) [41], a novel approach that leverages human-AI synergy in qualitative analysis. The CHALET pipeline consists of four key steps: (1) AI-assisted data collection through human-chatbot conversations, (2) human deductive coding to establish a baseline, (3) AI deductive coding using curated prompts, and (4) Human-AI coding **disagreements** analysis through collaborative inductive coding.

Our study demonstrated CHALET's effectiveness in advancing the interpretation of psychological constructs, particularly in the context of mental health stigma. I used the attribution model [15] as our theoretical underpinning. By analyzing human-AI coding disagreements, we uncovered several new themes and codes that extend beyond the original attribution model [15]. These findings deepen our understanding of stigma by incorporating more implicit stigmatizing thoughts beyond the attribution model, including microaggressions and other subtle attributions. For instance, we identified new cognitive judgments such as "self-neglect" and "vulnerability,"

emotional responses like "apathy" and "compassion fatigue," and behavioral responses including "differential support" and "paternalism." The CHALET approach also revealed psycholinguistic features that lead to human-AI discrepancies in coding mental health stigma, such as the use of distancing language and over-conjecture.

These insights enhance our understanding of how stigma manifests in language and pinpoint specific language patterns that impede AI interpretation, such as subtle emotional undertones and contextual nuances in human expression. Furthermore, our method demonstrated the value of human-AI collaboration by showing how AI can challenge human preconceptions and stimulate new analytical perspectives. For example, the AI's interpretation of certain expressions of *pity* as potentially stigmatizing prompted a reevaluation of this construct among human coders. Our approach provides a methodological framework for leveraging human-AI partnerships in theory-driven qualitative analysis of psychological constructs.

3.2 Study 2: AI for Explanatory Psychological-construct Analysis

While AI approaches to analyzing psychological constructs predominantly adopt descriptive coding methods [18], the complex interplay of contexts and reasoning chains within qualitative data often extends beyond what single labels can convey [61]. Recent research has shown that causal knowledge graphs excel at representing structured relationships between entities [9, 13], while LLMs demonstrate capabilities in semantic analysis and domain knowledge interpretation [29]. Inspired by this potential synergy [60], we developed an integrated pipeline for explanatory psychological-construct analysis. This pipeline consists of three main components: (1) AI-assisted data collection using chatbots to gather rich conversational data, (2) AI-assisted qualitative coding to identify key attributes of the psychological construct, and (3) causal knowledge graphs construction to map interrelationships among psychological factors. The causal knowledge graphs construction process involved triple extraction, ontologization, entity resolution, and conceptual model development, all supported by LLMs.

Our analysis produced a causal knowledge graph comprising 13,434 unique entities interlinked by 18,875 relationships, providing a rich network of constructs and causalities. First, we showed the ability to create participant-specific subgraphs, informing individual cognitive processes and allowing for tailored analysis of psychological constructs. Second, we uncovered novel constructs beyond the attribution model [15], including "personalities" and "past experiences." And third, we developed a conceptual model illustrating the drivers, triggers, and processes of stigma formation, integrating both established and newly discovered factors. This model reveals factor-pathway interplay, including how extrinsic factors like "situation" directly influence behavioral intentions, and how intrinsic factors like "personality" affect emotional responses.

Our approach demonstrated how LLMs and causal knowledge graphs can enhance the explanatory analysis of psychological constructs. The resulting analysis not only validated existing attribution theory but also resonated with social-cognitive theory's emphasis on how human behavior emerges from the interplay of individual characteristics and environmental contexts [3]. Additionally, our findings aligned with Lazarus' transactional model of appraisal [35], particularly in how cognitive judgments (i.e., primary appraisal) interact with personality traits and situational factors (i.e., secondary appraisal) to shape responses to mental illness.

4 Expected Next Steps, Open Questions and Challenges

4.1 Future Study 1: Interactive Visualization System for Exploring Psychological-construct Networks

To help HCI and social science practitioners better understand and utilize the causal relationships between psychological factors, I plan to develop an interactive visualization system for causal modeling exploration [64]. The system will enable users to explore our psychological networks with features for identifying central constructs through various centrality metrics (e.g., degree, betweenness, eigenvector centrality) [8], supported by dynamic exploration features like data tours, zoom controls, node filtering, and customizable layouts. Beyond exploration, practitioners can create their own knowledge structures by inputting raw data and theoretical frameworks - following our Study 2 methodology to establish causal models from their data while also being able to distill large-scale graphs into conceptual models. This dual functionality will support both the exploration of psychological causalities and theory extension, making complex psychological relationships more accessible and actionable for researchers and practitioners.

4.2 Future Study 2: Cross-cultural Analysis of Psychological Constructs through Multilingual AI

Our previous studies on psychological constructs have been limited by a participant pool primarily from WEIRD societies, potentially missing culture-specific nuances. Moreover, this limitation has hindered our ability to reflect on our own positionality as researchers when dissecting psychological constructs. To address these issues, I propose a cross-sociocultural comparative analysis focusing on

Singapore, Japan, China, and the U.S.A. This study will explore how culture intertwines with language in shaping psychological constructs. I plan to collect data from approximately 250-300 participants in each location, focusing on dialects and non-English languages such as Singlish, Japanese, and Chinese. Our method involves multilingual analysis of psychological constructs, followed by cross-sociocultural conceptual-model comparison. I will conduct linguistic analysis and topic modeling, and create interactive visualizations of factor-pathway mappings for different social groups. These tools will facilitate exploring cultural variations in psychological constructs, presenting complex relationships accessibly while prompting reflection on their own positionality. This study may face challenges in understanding and generating expressions in regional dialects. Accessing speakers of Asian languages and dialects and finding qualified annotators with both linguistic expertise and cultural understanding is inherently more difficult than for English. The performance and potential biases of LLMs in non-English contexts are also understudied, potentially adding another layer of complexity to the cross-cultural analysis.

4.3 Future Study 3: AI-enabled Tailored, Cultural Appropriate Psychological Intervention Design

Building on our cross-cultural insights, I aim to design tailored, culturally appropriate digital interventions using human-chatbot interactions. These interventions will incorporate various psychological techniques, including cognitive restructuring and motivational interviewing [24, 39]. Our theory-driven, personalized approach will be based on person-specific subgraphs, utilizing advanced chatbot design and retrieval-augmented generation techniques. The long-term nature of this study will enable us to observe and adapt to changes in user behavior and cognition over time. I will focus on developing a conversational agent that supports sustained engagement and facilitates natural, culturally appropriate interactions. Key challenges include incorporating established conversation norms for psychological intervention while maintaining user engagement over extended periods. The need for flexible content adjustment may lead to prolonged dialogues, potentially impacting user engagement. To address these challenges, I will explore multimodal delivery methods and integrate chatbot design elements such as active listening and restating techniques.

5 Research Status

I am a second-year PhD student in the AI for Social Good Lab at the School of Computing, Department of Computer Science, National University of Singapore. My advisor is Professor Yi-Chieh Lee. As part of the standard four-year PhD program at NUS, my expected completion date is June 2027. I have passed my qualifying examination. My long-term goal is to conduct research in human-centered AI (HCAI) and computational social science, exploring AI's potential for promoting cultural inclusivity and social justice in digital spaces. After graduation, I aim to pursue a career as an assistant professor in academia. I look forward to completing my thesis with feedback from the doctoral consortium, in which I have not yet participated in.

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